

Milestone 4 Report – Analysis-Ready Dataset Exploration

Project: Oncology Clinical Programming Portfolio

Study: Sanofi EFC10261 (NSCLC)

Objective

The objective of this milestone was to transform subject-level SDTM domains into an analysis-ready subject-level dataset by deriving exposure and disposition variables and integrating them with the Demographics (DM) domain. The work mirrors the process used to construct an ADSL dataset by creating reusable analysis components before assembling the final subject-level dataset.

Methods

Three modular programming workflows were implemented. First, the EX domain was summarized to one record per subject with treatment-specific and overall exposure timing variables. Second, the DS domain was summarized into subject-level disposition variables capturing end-of-treatment reason and last-contact information. Finally, DM was used as the subject-level backbone and the derived EX and DS components were integrated using RUSUBJID.

Analysis Components

EX Analysis Component:

- Derived first and last exposure study days for docetaxel and placebo.
- Calculated treatment-specific and overall treatment durations using inclusive study-day spans.

DS Analysis Component:

- Derived end-of-treatment reason.
- Derived last-contact status and retained last-contact study day where available.

ADSL Dataset:

- Combined subject demographics, exposure variables, and disposition variables into a single analysis-ready dataset.

Validation

Each analysis component underwent structural validation to confirm one record per subject and the absence of duplicate subject identifiers. Exposure derivations were checked for invalid durations and treatment end dates preceding treatment start dates. The final ADSL dataset was validated to ensure all DM subjects were retained and that exposure variables aligned with the study safety population.

Key Programming Decisions

- DM was used as the subject-level backbone.
- Overall treatment timing was derived exclusively from EX.
- End-of-treatment study day was intentionally not derived from SV because a disposition visit does not necessarily represent the actual final day of treatment exposure.
- Quality-control variables used during development were excluded from the final analysis datasets.

Results

The final ADSL dataset contained one record for each of the 455 subjects in the study. Exposure information was successfully integrated for all treated subjects, while missing values appropriately represented subjects without exposure or disposition records. Modular validation listings confirmed the integrity of the derived datasets.

Skills Demonstrated

- SDTM to analysis-ready dataset transformation
- Subject-level derivations
- Cross-domain integration
- Clinical data validation
- Reproducible R programming
- Modular workflow design